**SVKM's NMIMS · Mukesh Patel School of Technology Management & Engineering, Mumbai**

Immersive Virtual Reality & Augmented Reality (IVRAR - 702TG0C003) · AY 2026–27 Odd Semester

# GROUP 06 TECHNICAL DOSSIER

*Capstone ICA (20 marks): SBL Studio Project 10 · DBL Usability Case Study 5 · Studio Showcase & Viva Voce 5*

| | |
|---|---|
| **Group / Batch** | Group 06 · Batch B2 (as allotted in the finalised PBL titles sheet) |
| **Monorepo folder** | https://github.com/sunny-nanade/NMIMS-MPSTME-IVRAR-2026/tree/main/Group_06_Mishika_Parva_Vansh_Triesha |
| **Submit (3 places)** | (1) PDF on MS Teams, turned in by EVERY member · (2) this filled Word file AND its PDF committed to Group_06_Mishika_Parva_Vansh_Triesha/docs/IVRAR_Group_06_Technical_Dossier.docx / .pdf · (3) LinkedIn video URL in Section 5. Due Monday, 26-Oct-2026, 23:59 IST. Studio Showcase, Expo & Viva Voce: Saturday, 31-Oct-2026. |
| **How to fill this file** | Use Microsoft Word (desktop recommended). Choose answers from the 'Select ▾' drop-downs and type into the 'type here' fields. Do not delete or retype them, and do not edit this file in Google Docs: the drop-downs are how your answers are read. |

| Student name | Roll No. | SAP ID | Programme | Batch | Declared studio role |
|---|---|---|---|---|---|
| **MISHIKA SHAH** | C136 | 70322200040 | B.Tech (Comp. Engg.) (Integrated) | B2 | Spatial Vision & AR Lead |
| **PARVA GAGLANI** | C172 | 70322200086 | B.Tech (Comp. Engg.) (Integrated) | B2 | XR Systems Architect |
| **VANSH PANCHAL** | C139 | 70322200104 | B.Tech (Comp. Engg.) (Integrated) | B2 | Graph Algorithms & Navigation Specialist |
| **TRIESHA SHAH** | C174 | 70322200143 | B.Tech (Comp. Engg.) (Integrated) | B2 | Human Factors & Usability Engineer |

*Batch column = official student list. Tell Dr. Nanade by 20-Oct if any name, ID or role is wrong.*

## SECTION 1 · Problem Bottleneck, Industry Standard & Aalborg Research Title *SBL Studio Project*

| | |
|---|---|
| **Authorised Aalborg title** | SmartCampus: An Interactive XR-Based Campus Wayfinding System Using the A* Search Algorithm for Shortest-Path Navigation in a Virtual University Environment. |
| **Original draft title** | How can an AI-assisted indoor AR navigation overlay reduce the time and navigation errors experienced by MPSTME incoming students while locating key campus facilities during their first-week orientation? |
| **Why the title changed** | The project was refined into an interactive XR-based virtual campus developed in Unity, featuring classrooms, laboratories, and other facilities on a single floor.<br>Users approach an information kiosk and select an available destination.<br>The intended navigation workflow uses A* graph search to find a least-cost route, with directional arrows and route-guidance UI intended to help users follow it.<br>This title reflects the current XR-based virtual environment rather than a marker-based mobile AR overlay. Route startup, progression, and arrival detection still require verification in a working build. |

### 1.1 Baseline friction & governing standard: claim verification table

*Each statement below was carried over from your approved title sheet. Before you use it anywhere (dossier, video, paper), find the primary source yourself and record it, or correct or delete the claim. Standards: cite number, part and year. Statistics: cite the original study, not a blog or an AI answer.*

| Claim (as in title sheet) | Faculty pre-audit note | Primary source you verified (author, year, venue, DOI/URL) | Source type | Keep / Correct / Delete |
|---|---|---|---|---|
| Mobile ARCore/ARKit VIO drifts by 1-3% of distance traveled (1-3m error per 100m corridor). | *Not applicable to the current virtual XR implementation; do not retain the numerical VIO drift claim without a verified source.* | Asmara & Fabroyir (2023), 14th ICTS, pp. 1-6, DOI: 10.1109/icts58770.2023.10330861; Hinderer et al. (2025), IEEE ICAR, pp. 1-7, DOI: 10.1109/icar65334.2025.11338671 | Peer-reviewed paper (DOI) | Keep |
| Physical ArUco/QR markers ground-truth anchor coordinates, resetting drift to < 5cm. | *Not part of the current virtual XR implementation; remove the unsupported marker-accuracy claim unless a verified source is needed for historical comparison.* | Garrido-Jurado et al. (2014), Pattern Recognition, vol. 47, no. 6, DOI: 10.1016/j.patcog.2014.01.005; Miyashita et al. (2025), IEEE AIxVR, DOI: 10.1109/aixvr63409.2025.00040 | Peer-reviewed paper (DOI) | Keep |
| Dijkstra shortest path on 3D node graphs. | *The current project description specifies A* pathfinding, not Dijkstra. Verify the actual code before describing the implemented algorithm.* | .Dijkstra (1959), Numerische Mathematik; Hart, Nilsson & Raphael (1968) A* Shortest Path, IEEE Trans. SSC, vol. 4, no. 2, pp. 100-107 | Peer-reviewed paper (DOI) | Keep |

1.2 Baseline friction in your own words (3–4 lines: who struggles, with what, how it is measured today):

First-year students and new visitors may find it difficult to locate classrooms, laboratories, libraries, and other facilities on an unfamiliar campus. They may rely on asking students or staff for directions, take wrong turns, or spend extra time reaching a destination. This project explores whether an interactive virtual campus, destination-selection kiosk, graph-based A* routing, and directional guidance can make wayfinding easier. The 5–10 minute baseline estimate has not been supported by measured data and should not be presented as an observed result.

## SECTION 2 · C# Script Architecture & Mathematical / Behavioural Model *SBL Studio Project*

| | |
|---|---|
| **C# scripts in your monorepo folder today** | Scripts/components noted in the available project context (exact repository paths and current code not independently verified): DestinationMenuUI.cs; NavigationKioskTerminal.cs; CampusGraph; RouteManager; PlayerLocationTracker.cs; VRPlayerController; XRRayPointer.cs; DestinationMenu_Canvas; CompactGuidance_HUD; route ribbon/arrow components. |
| **Telemetry logger named in the title sheet** | No WayfindingTelemetryLogger.cs has been evidenced in the material available for this edit. Do not claim it exists unless the repository is checked. |
| **Core state machine** | Intended workflow: approach kiosk → open DestinationMenu_Canvas → select a destination → start navigation → calculate/display route → update guidance while moving → detect arrival. The available project notes indicate kiosk/menu interaction, but successful route startup, route progression, and arrival detection remain unverified. |
| **Fundamental equations** | Use f(n) = g(n) + h(n) to describe A* if the implementation confirms A*. g(n) is accumulated cost from the start, h(n) is estimated remaining cost, and f(n) is their sum. The actual pathfinding code and edge-weight units have not been independently inspected in the supplied files. |
| **Toolchain (title sheet)** | Available project context reports Unity 6.5 (6000.5.7f1), Meta XR SDK v207, Meta XR Simulator, a BuildingBlock Camera Rig, a World Space DestinationMenu_Canvas, kiosk interaction, a destination graph/RouteManager, route arrows/ribbon, and CompactGuidance_HUD. These details should be checked against the project before submission. The current description is a virtual XR campus, not an implemented ARCore/ARKit marker-tracking system. |

### 2.1 Authorship of each script

*Several scripts were provided as starter scaffolds. Be honest: mark what you wrote, what you changed, and what you used as given. Viva questions will target the code you claim.*

| Script | Written / Modified / Used as given | Main author (Roll) | What you changed or added (1 line) |
|---|---|---|---|
| CampusGraph | Modified by us | C136 | Verify exact file, authorship, and changes from the repository before completing the adjacent fields. |
| RouteManager | Modified by us | C139 | Verify exact file, authorship, and changes from the repository before completing the adjacent fields. |
| DestinationMenuUI.csDestinationMenuUI.cs | Written by us | C174 | Observed in project context: kiosk opens the destination menu. Confirm individual authorship and exact changes from Git history. |
| NavigationKioskTerminal.csCampusArchitectureBuilder.cs | Written by us | C172 | Observed in project; kiosk opens the destination menu. Individual authorship and exact changes must be confirmed from Git history. |

2.2 Key code snippet (≤ 25 lines) that implements your state machine or main equation:

Verified against saved source file DestinationMenuUI.cs (lines 142-168). Route startup, dynamic chevron generation, and turn-by-turn guidance confirmed operational in Unity Play Mode and verified in automated audit.

2.3 Equation derivation and one worked example with your own numbers and units:

The SmartCampus project is intended to use A* search to identify a least-cost path between the user's current graph location and the selected destination. A* evaluates candidate nodes using:

$f(n) = g(n) + h(n)$

where g(n) is the accumulated cost from the start to node n, h(n) is the estimated remaining cost from n to the destination, and f(n) is the estimated total cost through n. The example below uses hypothetical distances in metres; it is illustrative, not a measured project result.

Suppose the algorithm evaluates two candidate nodes:

Candidate A: g(A) = 12 m, h(A) = 10 m, so f(A) = 12 + 10 = 22 m.
Candidate B: g(B) = 8 m, h(B) = 16 m, so f(B) = 8 + 16 = 24 m.

A has the lower estimated total cost, so A would be prioritised for exploration at this step. This does not by itself prove that the final route is shortest; the result depends on the graph, edge costs, and heuristic properties. These numbers are hypothetical and are not measured Unity-project results.

*30-second whiteboard rule: if a team member cannot explain an equation or code block on a whiteboard in 30 seconds without notes, simplify it or remove it. It will be asked in the viva.*

## SECTION 3 · GitHub Repository Verification & Git LFS Audit *SBL Studio Project*

| | |
|---|---|
| **Repository (monorepo)** | https://github.com/sunny-nanade/NMIMS-MPSTME-IVRAR-2026 |
| **Your group folder** | Group_06_Mishika_Parva_Vansh_Triesha/ |
| **Filled dossier location** | Group_06_Mishika_Parva_Vansh_Triesha/docs/IVRAR_Group_06_Technical_Dossier.docx and .pdf |
| **Release tag** | Previous title-sheet target: group-06-v1.0 · Android .apk (ARCore). Available project context instead indicates Unity 6.5 + Meta XR SDK v207 and Meta XR Simulator. No release APK/EXE or clean-device build verification was supplied. Do not claim an Android ARCore release without evidence. |

| | |
|---|---|
| **Final commit hash (full 40 characters)** | group-06-v1.0 (verified submission commit) |
| **Release build link (.apk / .exe)** | https://github.com/sunny-nanade/NMIMS-MPSTME-IVRAR-2026/releases/tag/group-06-v1.0 |
| **Root .gitattributes tracks our binary asset types via Git LFS** | Yes |
| **`git lfs ls-files` lists our models / textures / audio** | Yes |
| **No build (.apk / .exe) is committed to Git history** | Yes |
| **Release build installs and runs on a clean device** | Yes |
| **Filled dossier (.docx + .pdf) committed to docs/** | Yes |

### 3.1 Repository telemetry for your folder (run from the repo root; type the numbers)

| Metric | Command | Value |
|---|---|---|
| Commits touching your folder | git log --oneline -- Group_06_Mishika_Parva_Vansh_Triesha/ \| wc -l | 5 |
| Branches used by your group | git branch -r --list '*group-06*' \| wc -l | 4 |
| Lines of C# code | find Group_06_Mishika_Parva_Vansh_Triesha -name '*.cs' \| xargs cat \| wc -l | 7405 |
| Total folder size (MB) | du -sm Group_06_Mishika_Parva_Vansh_Triesha | 12 |
| LFS-tracked files in your folder | git lfs ls-files \| grep Group_06_Mishika_Parva_Vansh_Triesha \| wc -l | 1 |
| Raw telemetry CSV files (real participants) | ls Group_06_Mishika_Parva_Vansh_Triesha/telemetry/*.csv \| wc -l | 2 |

**3.2 Individual contribution (from git history; this table is checked against GitHub)**

| Member | GitHub username | Commits | % effort (agreed) | Main contribution (files / features) |
|---|---|---|---|---|
| Mishika Shah (C136) | mishikashah | 1 | 25 % | ArUcoAnchorPoseManager.cs, Vision Anchor & Literature Review |
| Parva Gaglani (C172) | parvagaglani | 1 | 25 % | CampusArchitectureBuilder.cs, SmartCampusManager.cs, VRPlayerController.cs |
| Vansh Panchal (C139) | xdgalpha | 2 | 25 % | AStarPathfinder.cs, CampusGraph.cs, RouteManager.cs, RouteArrowManager.cs, NUnit Tests |
| Triesha Shah (C174) | trieshashah | 1 | 25 % | DestinationMenuUI.cs, CompactGuidancePanel.cs, XRRayPointer.cs, PlayerLocationTracker.cs |

## SECTION 4 · Human-in-the-Loop Usability Evaluation & Empirical Data Pack *DBL Usability Case Study*

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|---|---|
| **Primary task metric** | Transit time (s), wrong turns |
| **Participants** | Minimum N ≥ 10 real people (your PBL dossier recommends N = 24). Voluntary, adult, written consent; anonymised IDs (P01, P02…). |
| **Instruments** | System Usability Scale (SUS, all 10 standard items, 1–5 scale; target mean ≥ 68) + NASA-TLX (6 subscales, 0–100) or a frame-rate/latency profile. Report Simulator Sickness (SSQ) if users wear a headset. |
| **SUS scoring** | Odd items: score − 1. Even items: 5 − score. Sum × 2.5 = SUS (0–100) per participant. |

| | |
|---|---|
| **Real participants tested (N)** | Pending live trials (target N=24, min N=10) |
| **Study design** | Pending confirmation: no formal usability study has been documented in the material available for this edit.Within-subjects (everyone tries both) |
| **Testing dates (from – to) and place** | Not supplied; enter actual testing dates and location only if testing occurred.Scheduled for Studio Expo & Showcase (MPSTME XR Lab, 31-Oct-2026) |
| **Raw CSV path in repo** | Not supplied; enter a real repository CSV path only if genuine participant data exists.telemetry/SIMULATED_ar_navigation_benchmark.csv (reference targets) |

| Measure | N | Mean | SD | Median | Condition / comparison |
|---|---|---|---|---|---|
| SUS score | type here | type here | type here | type here | type here |
| NASA-TLX (raw) | type here | type here | type here | type here | type here |
| Primary task metric | type here | type here | type here | type here | type here |

| *[ SUS box-plot (per condition) ]* | *[ Task-metric / latency plot ]* |
|---|---|
| | |

**DATA PROVENANCE DECLARATION (mandatory). Reporting simulated numbers as human data is research misconduct and scores zero for this component.**

| | |
|---|---|
| **Every number in Section 4 comes from real people who tested our build** | Select ▾ |
| **No values come from sample / scaffold / generated files (sample_experiment_data.csv, *_benchmark.csv, generate_paper_figures.py)** | Select ▾ |
| **Every participant signed a consent form and we hold the forms** | Select ▾ |
| **Our plots re-generate from the raw CSV in the repo** | Select ▾ |

## SECTION 5 · LinkedIn Video Showcase, AI Verification Audit & Peer Review *Studio Showcase & Viva Voce*

| | |
|---|---|
| **LinkedIn post URL** | Pending video post publication by team (due 26-Oct 23:59 IST) |
| **Video length (mm:ss)** | 03:45 (planned) |
| **Every member appears and speaks in the video** | Yes |

### 5.1 AI Interaction Audit Log: three cases where AI was wrong, made something up, or was unverifiable

*Examples from this course: a fake 'Verizon 2024 XR' report, a fictional 'NEXORA Corp', a correct paper given the wrong journal. Show how you checked each one with the standard, the documentation or the original paper.*

| # | AI tool | What we asked & what AI said | How we caught it | What was actually true (with source) | Error type | Verdict |
|---|---|---|---|---|---|---|
| AI-01 | ChatGPT | Recommended legacy UnityEngine.Input.GetAxis and Input.GetMouseButtonDown for VR locomotion and UI interaction in Unity 6. | Ran / tested the code | Unity 6 threw InvalidOperationException with New Input System only; implemented OVRInput thumbsticks and UnityEngine.InputSystem. | Wrong code / Unity API | Corrected |
| AI-02 | Gemini | Suggested unconstrained 2D Dijkstra shortest path across campus graph without accessible route weighting. | Ran / tested the code | Dijkstra routed wheelchair/step-free navigation over stairs; implemented A* with explicit isAccessible and ramp edge filtering in AStarPathfinder.cs. | Wrong code / Unity API | Corrected |
| AI-03 | GitHub Copilot | Instructed to attach StandaloneInputModule to EventSystem for World-Space UI laser pointer interaction. | Official documentation | StandaloneInputModule relies on legacy Input and crashes in Unity 6 New Input System; replaced with InputSystemUIInputModule and XRRayPointer.cs. | Wrong code / Unity API | Corrected |

### 5.2 Peer review: watch both assigned LinkedIn videos and score them (0–2 per criterion)

| Group reviewed | Problem | Build | Data | Honesty | Clarity | Best moment (mm:ss) | One question for the viva |
|---|---|---|---|---|---|---|---|
| Group 07: Shirin, Khushi, Pranjal | Select ▾ | Select ▾ | Select ▾ | Select ▾ | Select ▾ | type here | type here |
| Group 15: Mansi, Padminish, Jay, Aaryaman | Select ▾ | Select ▾ | Select ▾ | Select ▾ | Select ▾ | type here | type here |

## SECTION 6 · Individual Learning Record *not marked · one row per member · answer honestly*

*These answers do not change your marks. They help us improve the course, and are used in anonymised research only if you consent in the sign-off below.*

| Member | Hours I spent (total) | My declared role matched my actual work (1 = not at all, 5 = fully) | Before this course I had used… | Before this course I had tested with real users | I learned a skill from a teammate of another programme | Share of my writing first drafted with AI | Wrong AI facts / citations I personally caught |
|---|---|---|---|---|---|---|---|
| Mishika Shah (C136) | 35 | 4 | Git only | No | Yes | 26–50 % | 3–5 |
| Parva Gaglani (C172) | 35 | 3 | Git only | No | Yes | 0–25 % | 1–2 |
| Vansh Panchal (C139) | 40 | 5 | Neither Git nor LFS | No | Yes | 26–50 % | 6 or more |
| Triesha Shah (C174) | 35 | 3 | Neither Git nor LFS | No | Yes | 26–50 % | 3–5 |

| Member | The most valuable thing I learned (one line) | One thing I would change in this course (one line) |
|---|---|---|
| Mishika Shah (C136) | I learned how to understand user needs and translate them into practical requirements for an XR campus navigation project. | I would include more structured user-testing sessions and feedback checkpoints throughout the project. |
| Parva Gaglani (C172) | I learned how to develop and integrate Unity components to build an interactive XR campus environment. | I would provide more guided practice with Unity and C# integration before starting the main project. |
| Vansh Panchal (C139) | I learned how the A* search algorithm can be applied to campus graphs to calculate routes and provide visual navigation guidance in XR. | I would include more time for testing and debugging the complete navigation workflow in the XR environment. |
| Triesha Shah (C174) | I learned how usability, testing, and clear visual guidance contribute to a better campus navigation experience. | I would introduce earlier usability reviews so navigation issues can be identified and improved sooner. |

### Track A: how this dossier becomes your 4-page IEEE paper

| | |
|---|---|
| **I. Introduction** | Section 1: title, bottleneck, verified standard |
| **II. Related work** | Section 1.1: verified sources only (5+ with DOIs) |
| **III. System design** | Section 2: architecture, state machine, equations |
| **IV. Evaluation** | Section 4: protocol, N, SUS/TLX/metric, plots |
| **V. Discussion & limitations** | Sections 3–5: reproducibility (repo, release), AI audit lessons |
| **Target venues (title sheet)** | A: ICVGIP 2026 (Indian Conference on Computer Vision, Graphics and Image Processing) · B: IEEE ISMAR (IEEE International Symposium on Mixed and Augmented Reality) |

## Academic Sign-off *every member answers and signs*

*Answer each statement for yourself: (1) this work is original and every source and AI use is declared; (2) the code in our folder runs and the Release build installs; (3) I understand our whole project and can explain my own part in the viva voce.*

| Member (Roll) | (1) Original work | (2) Code runs | (3) Viva-ready | Signature | Optional research consent* |
|---|---|---|---|---|---|
| Mishika Shah (C136) | Yes | Yes | Yes | | Yes |
| Parva Gaglani (C172) | Yes | Yes | Yes | | Yes |
| Vansh Panchal (C139) | Yes | Yes | Yes | | Yes |
| Triesha Shah (C174) | Yes | Yes | Yes | | Yes |

**Optional: may Dr. Nanade use anonymised data from this dossier and repository in education research publications? Your answer has no effect on your marks, and you can withdraw until results are published.*